Southwire®
POWER GENERATION SOLUTIONS

SK-3X1 POWER CAMPUS

STRATEGY MONTHLY UPDATE

EVERY MEGAWATT TRAVELS THROUGH A CABLE



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ONE CAMPUS.
EVERY CABLE.

A 3x1 combined-cycle plant with RICE engines, carbon capture, battery storage, a modular power yard, an LNG terminal and a behind-the-meter data centre: one campus, and from the 230 kV grid connection to the last instrument loop, all of it runs on cable. The next slides follow that cable through the plant.

01	GRID CONNECTION	SLIDE 04
02	POWER ISLAND · TURBINE HALL	SLIDE 06
03	CARBON CAPTURE	SLIDE 09
04	BATTERY STORAGE	SLIDE 10
05	MODULAR & TEMPORARY POWER	SLIDE 11
06	INSTALLED FASTER	SLIDE 12



THE PLANT AT A GLANCE

WHERE THE CABLE GOES.

Take the steel away and this is what is left: one network, from the 230 kV grid connection to the last instrument loop, in every zone of the campus.

BY THE NUMBERS

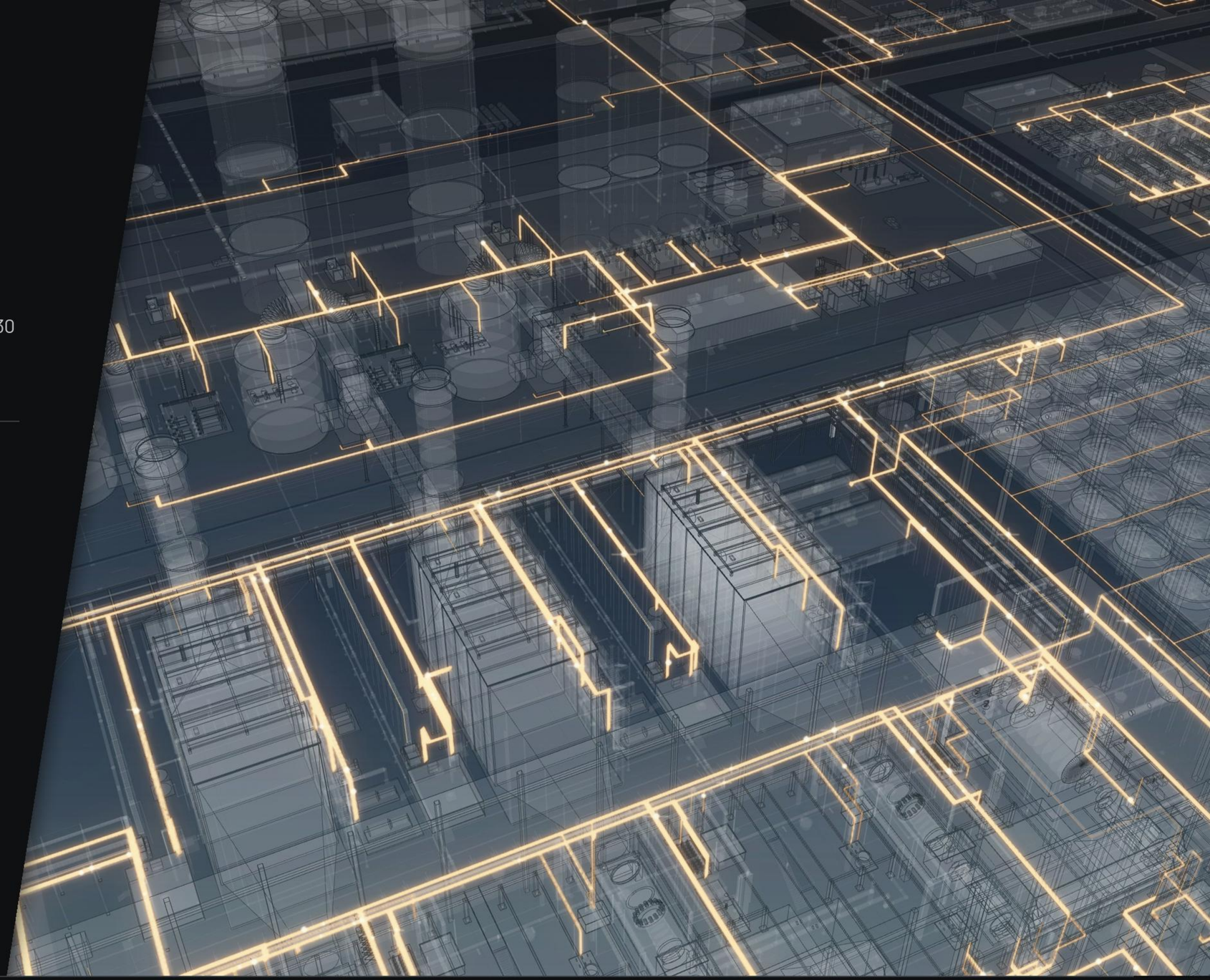
4.6 MI
of cable tray routes on the racks and in the halls

4.9 MI
of duct-bank and buried cable routes

4,946 FT
of 230 kV underground cable routes

418
pieces of equipment with their own cable applications

Route lengths measured along the modelled tray, duct-bank and cable routes (not cable footage); data-centre campus excluded. Picture: the 3D model in cable X-ray.



230 kV TO THE GRID

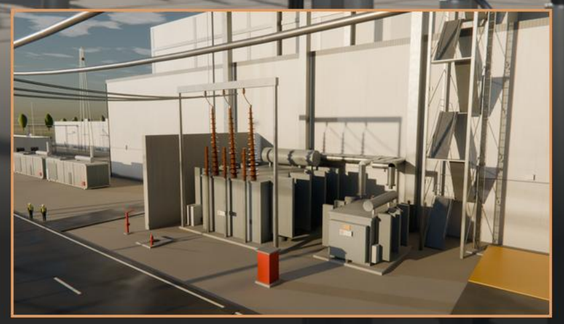
The plant's output leaves through four generator step-up transformers and a breaker-and-a-half 230 kV switchyard. Where the line runs underground, high-voltage XLPE cable rises on cleated risers to sealing ends and surge arresters.

01 HV XLPE TRANSMISSION CABLE
Copper conductor, XLPE insulation, aluminium laminate moisture barrier and HDPE jacket for the buried 230 kV run.

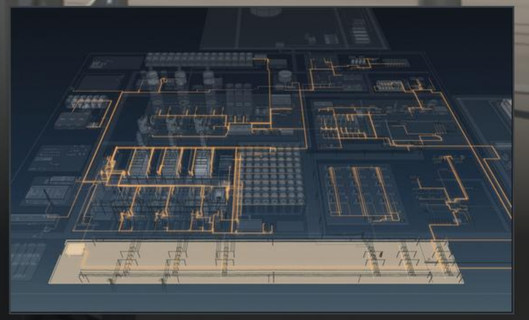
02 FEWER JOINTS
Long continuous lengths keep splices out of the duct bank.

03 ONE SUPPLIER
Cable, accessories and field support for the complete circuit.

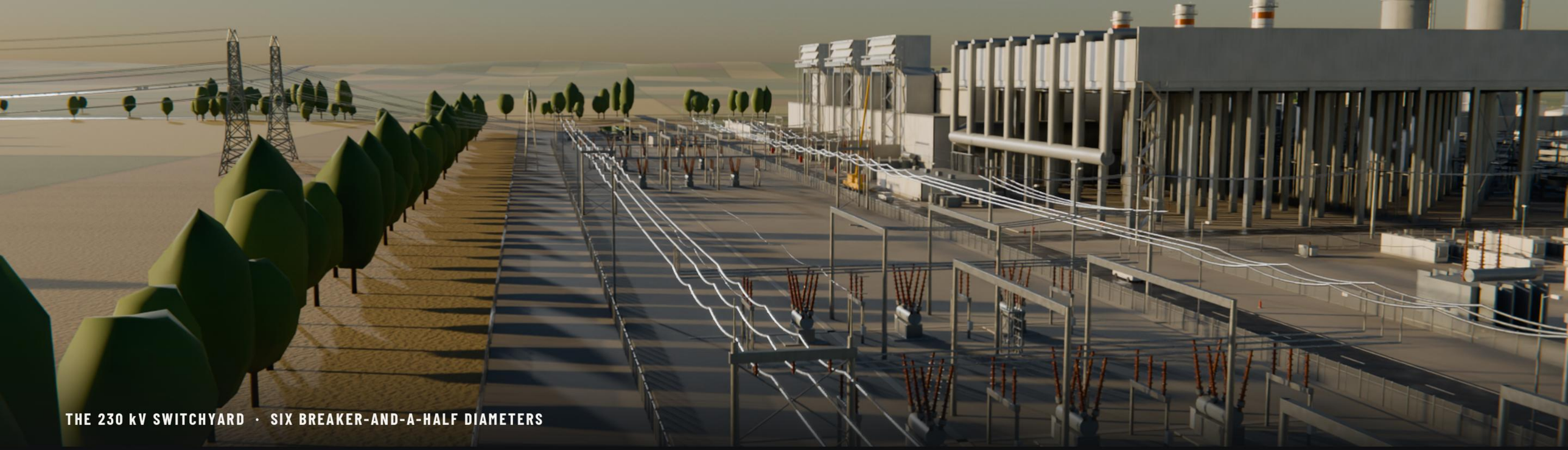
230 kV | 2500 KCMIL CU | XLPE | AL LAMINATE | ICEA S-108-720 | 4/0 BARE CU GROUND



Generator step-up transformers facing the switchyard



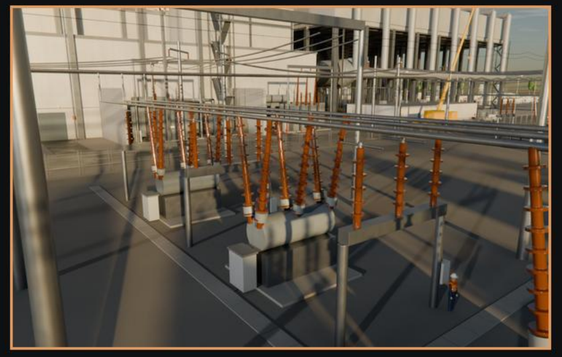
WHERE IN THE PLANT



THE 230 kV SWITCHYARD · SIX BREAKER-AND-A-HALF DIAMETERS



Along the switchyard at eye level



Bay D2: breakers, disconnects, trench to the relay house

WHAT RUNS AT THE GRID CONNECTION

HV XLPE 230 kV

1/C 2500 KCMIL CU · XLPE · CU WIRE SHIELD · AL LAMINATE · HDPE

Underground 230 kV circuits and the CCS / modular-yard ties

BARE COPPER

4/0 AWG GROUND GRID · RODS · RISERS

Station ground grid under the yard and every structure

TYPE TC-ER CONTROL

600 V MULTICONDUCTOR · EXPOSED-RUN RATED

Breaker and relay circuits in the yard trenches

MV-105 35 kV

1/C 500 KCMIL AL · EPR 133% · 1/3 CN · LLDPE

Station service and collector feeders

ARMOR-X
IN THE TRAY

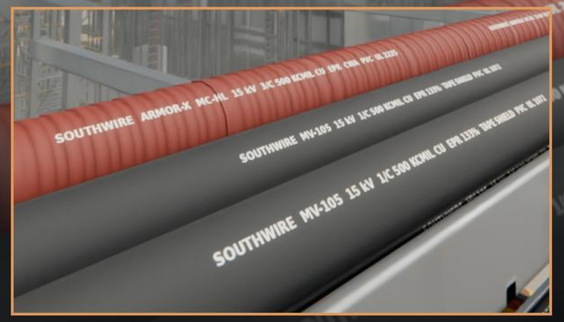
Three H-class gas turbines and a steam turbine share one hall. 15 kV ARMOR-X MC-HL and MV-105 run from the 13.8 kV switchgear across the tray stack to every auxiliary drive, with 600 V ARMOR-X and Type TC-ER for power and control in the gas-turbine enclosures.

01 HAZARDOUS-LOCATION RATED
MC-HL: continuously welded, corrugated aluminium armor under a PVC jacket, for the Class I, Div. 2 areas around the gas turbines.

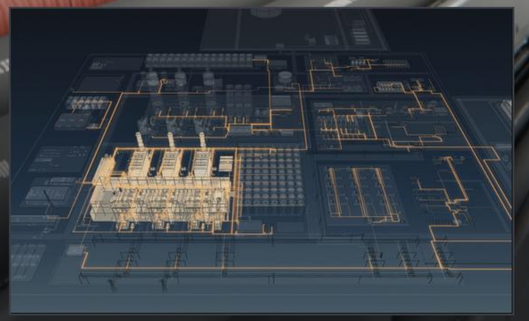
02 NO CONDUIT
Armored cable goes straight into the tray: fewer trades, faster pulls.

03 READABLE EVERYWHERE
Printed legend on the jacket for identification at every drop.

15 kV MV-105 133% · RED PVC | 600 V CSA RA90-HL · BLACK PVC | UL 1569 / UL 2225 | TYPE TC-ER | MV-105 · UL 1072



The legend on every jacket: ARMOR-X MC-HL 15 kV beside MV-105


WHERE IN THE PLANT

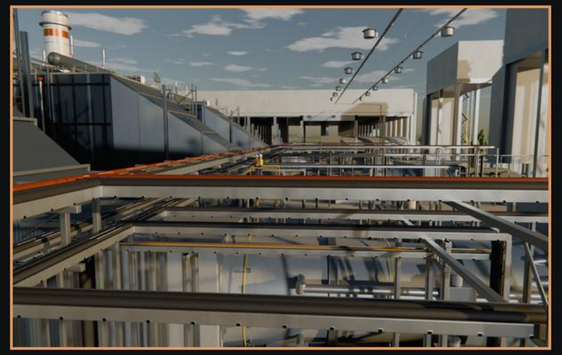
| STEPHAN HARDT stephan.hardt@southwire.com
Product data typical; confirm against current Southwire specifications.



OUTAGE PULL · NEW 13.8 kV FEEDER FROM THE LAYDOWN BAY UP INTO THE TRAY



ARMOR-X and MV-105 in the EL 48 tray over GT1



Tray stack along the hall north wall, drops to the GT skids

WHAT RUNS IN THE TURBINE HALL

ARMOR-X® MC-HL 15 kV

500 KCMIL 3/C CU · NL-EPR 25% TAPE SHIELD · GROUND · RED PVC

MV feeders to auxiliary drives through the gas-turbine areas

ARMOR-X® MC-HL 600 V

#14 AWG TO 750 KCMIL · XHHW-2 · CSA RA90-HL · BLACK PVC

Motor and power circuits on the enclosures and skids

MV-105 15 kV 133%

1/C 500 KCMIL CU · EPR · TAPE SHIELD · PVC · UL 1072

Switchgear to transformers and large motors

TYPE TC-ER

600 V POWER AND CONTROL · EXPOSED-RUN RATED

Control, protection and instrument circuits along the tray

THREE TURBINES. ONE TRAY STACK.

Three H-class gas turbines and the steam turbine share one hall. Every auxiliary drive, enclosure fan, lube-oil skid and protection relay is fed from the tray stack that runs the length of the hall at EL 30 to 48: medium voltage on top, 600 V power below, control and instrumentation at the bottom.

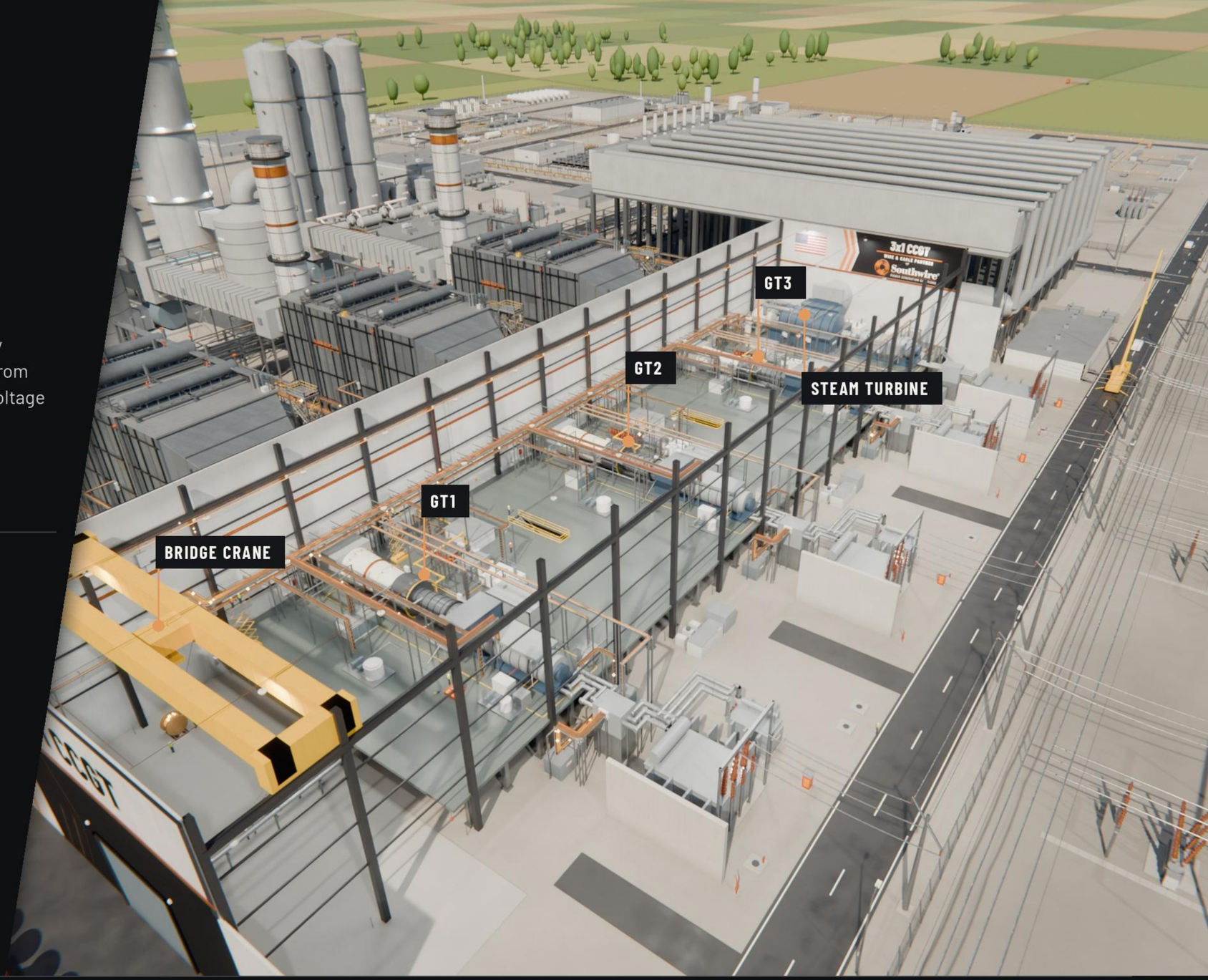
THE HALL BY THE NUMBERS

5,604 FT
of cable tray inside the hall

1,047 FT
of it 15 kV medium-voltage tray

208 FT
of isolated-phase bus, generators to GSUs

61
pieces of equipment in the hall with their own cable



Lengths measured along the modelled routes inside the hall.

CABLE THAT RUNS CCS

Absorbers, strippers and 16 MW booster fans: post-combustion capture adds a second plant's worth of motors, drives and instruments, all fed from the CCS rack.

01 ARMOR-X ON THE RACK
Jacketed armor for the process area's power circuits.

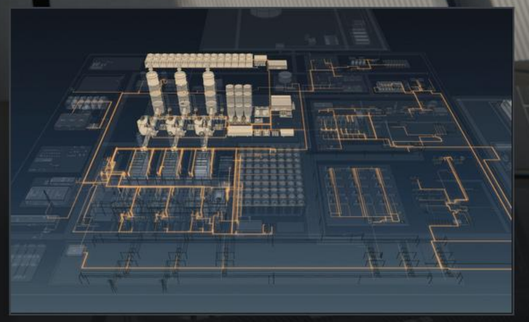
02 15 KV FEEDERS
MV-105 to the fans and the CO2 compressors.

03 INSTRUMENTATION
PLTC and Type TC-ER for the absorber and regeneration loops.

230 kV XLPE | 15 kV MV-105 | 600 V ARMOR-X | TYPE TC-ER | PLTC



CCS transformers T-1 / T-2: 230 kV terminations and the tray riser



WHERE IN THE PLANT

DC IN THE TRENCH

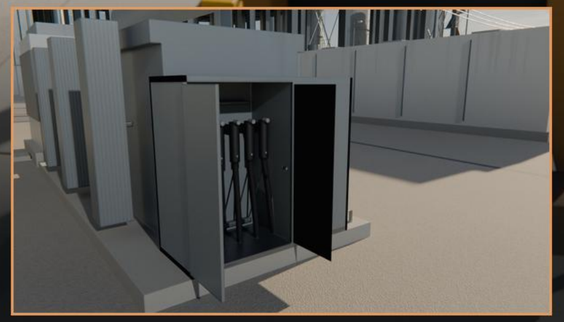
The battery yard pairs containers with power-conversion skids. 1,500 V DC conductors run in precast trenches; 35 kV MV-105 collects the skids to the BESS substation.

01 PV / RHW-2
Sunlight- and moisture-resistant 2 kV conductors for the DC side.

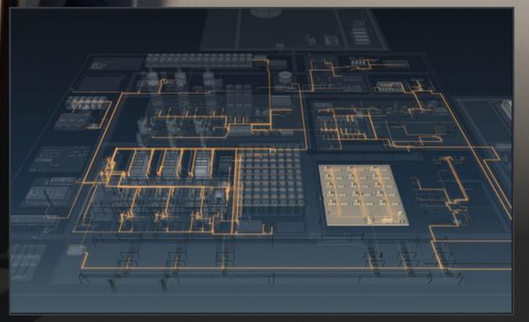
02 MV COLLECTION
35 kV MV-105 loop feeds with load-break elbows.

03 PULLED ON SITE
Crews pay off from reels straight into the open trench.

2 kV PV / RHW-2 | UL 4703 | 35 kV MV-105 | UL 1072



PCS / MV skid: 34.5 kV elbow terminations to the collector



WHERE IN THE PLANT

POWER THAT MOVES

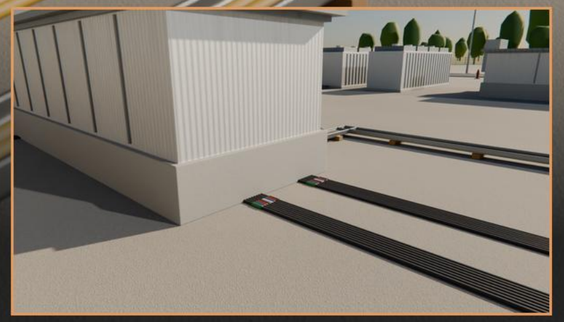
Gensets, trailer turbines and bridge-power fleets come and go. Portable power cable lies on the pad, crosses lanes under drive-over protectors and plugs in with cam-locks; bridge units feed through ground trays to the permanent duct bank.

01 TYPE SHD-GC
Flexible 15 kV portable power cable for the MV feeders.

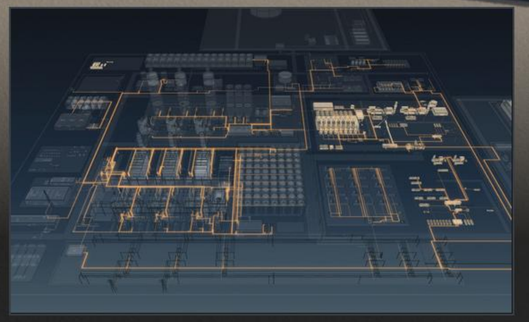
02 TYPE W
2000 V single conductors with cam-lock connections for 480 V hookups.

03 MV-105 LEADS
Generator leads from the TM2500 terminal boxes in ground trays.

15 kV SHD-GC | ICEA S-75-381 | 2000 V TYPE W | CAM-LOCK | 15 kV MV-105



Type W sets with cam-lock plugs into the 480 V switchboard



WHERE IN THE PLANT

ACCELERATING TIME TO POWER™

Schedule is decided in the field. The SIMpull Truck delivers multi-conductor SIMpull Reels and pays them off at the pull, so a feeder goes in with one setup instead of three.

01 SIMPULL REELS
Paralleled conductors on one reel: one pull per feeder.

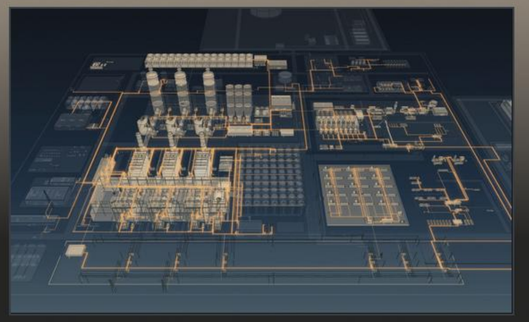
02 SIMPULL TRUCK
Delivery, payoff and reel return in one vehicle.

03 FEWER SETUPS
Less reel handling on site, faster energization.

SIMPULL® REELS | SIMPULL TRUCK | JOBSITE SERVICES



The cable reel yard: SIMpull Reels staged for the pulls



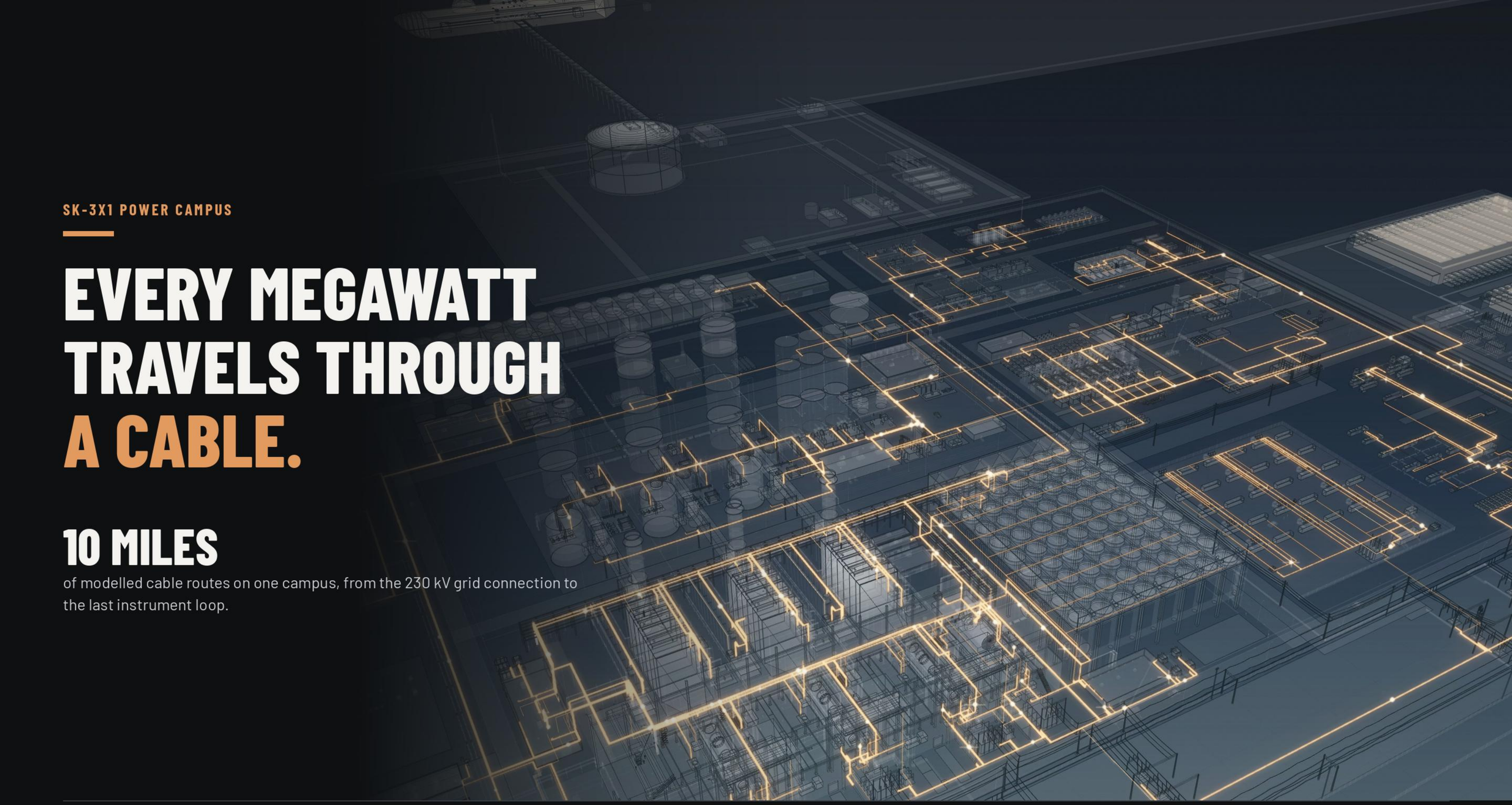
WHERE IN THE PLANT

SK-3X1 POWER CAMPUS

EVERY MEGAWATT TRAVELS THROUGH A CABLE.

10 MILES

of modelled cable routes on one campus, from the 230 kV grid connection to the last instrument loop.



PRODUCT INDEX

ONE SUPPLIER, GRID TO STACK

ZONE	PRODUCT	VOLTAGE	STANDARD
GRID CONNECTION	HV XLPE transmission cable	230 kV	ICEA S-108-720
	Bare copper grounding	4/0 AWG	—
POWER ISLAND	ARMOR-X® MC-HL, red PVC	15 kV	UL 1569 / 2225
	ARMOR-X® MC-HL, black PVC, CSA RA90-HL	600 V	UL 1569 / 2225
	MV-105 EPR 133%	15 kV	UL 1072
	Type TC-ER power and control	600 V	UL 1277
CARBON CAPTURE	ARMOR-X, MV-105, TC-ER, PLTC	300 V – 15 kV	UL 13 / 1072 / 1277
BATTERY STORAGE	PV / RHW-2 DC conductor	2 kV	UL 4703
	MV-105 collection	35 kV	UL 1072
MODULAR POWER	Type SHD-GC portable power cable	15 kV	ICEA S-75-381
	Type W portable power cable	2000 V	UL 1650
INSTALLATION	SIMpull® Reels and SIMpull Truck	—	—

Southwire®
POWER GENERATION SOLUTIONS

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